University of Glasgow | School of Computing Science

Software Engineering Placement, COMPSCI4046
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Keeping Misrata Connected

A redesigned customer portal and SMS outage alerts for an ISP

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THE 45 SECOND VERSION

LT NET's monitoring could **see a tower go down but could not tell anyone**. Its customer portal was dated and not built for Arabic. In nine weeks I **redesigned the portal** and built **SMS alerts for engineers**.

About LT NET

An internet service provider in **Misrata, Libya**. It sells home and business connections over physical infrastructure spread across the city. I worked alongside the **software and network engineers**, building for two groups: customers and engineers.

~20	9	2
towers	weeks	projects

Company size: [approx. staff / customers]

Two gaps, two problems

FOR CUSTOMERS

A dated portal

Built on the DMA Radius Manager billing system. Hard to use, minimal branding, and the layout was **not designed to read right to left** for Arabic.

FOR ENGINEERS

Silent outages

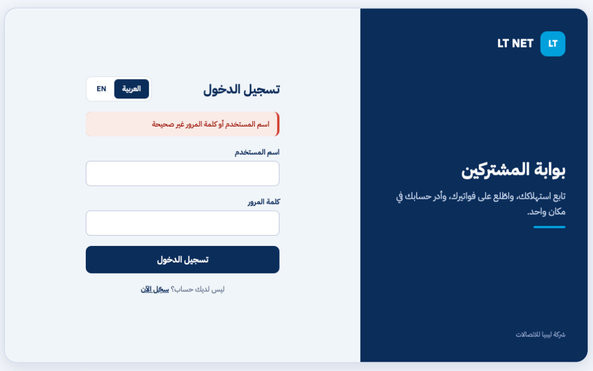
The monitoring system checked if each tower was up or down, but it had **no way to tell anyone**. Someone had to watch the screen.

1 Redesigning the customer portal

Plain PHP, HTML, CSS and JavaScript on DMA's existing API



BEFORE **DMA default login**



AFTER **Branded, bilingual login**



AFTER **Dashboard (demo account)**

RIGHT TO LEFT
Layout properly mirrored for Arabic

BILINGUAL
AR / EN toggle on every screen

BRANDING
Layout, colour and font agreed with the design team

SELF SERVICE
Balance, usage, invoices and connection in one view

The constraint: LT NET set the scope as a frontend redesign. DMA generates its own customer pages and is locked down, so I **reused its existing API** instead of building a new one.

2 Silent outages to SMS alerts

Python cron job, REST call to Madar / Libyana, SMS to engineers

Monitoring database
Up / down status per tower

↓

Cron job
Python script runs on a schedule

↓

REST call
Madar / Libyana SMS gateway

↓

SMS to engineer: tower name + time

Recipient numbers live in a database, so adding an engineer never touches the script.

1. Tower goes down
One SMS, then the tower is flagged so it does not fire every run.

2. Tower recovers
A "back up" SMS is sent and the flag is cleared.

3. Send fails
Only an HTTP 200 marks it alerted, so failures retry automatically.

STATUS Built and in testing with LT NET engineers. Not yet used by customers.

Tools and practices

PHP · HTML · CSS · JavaScript · Python · requests · cron · REST API · DMA Radius Manager · Madar / Libyana SMS

- Reused the existing API inside a locked system
- Design reviewed with LT NET's design team
- State tracking to avoid alert spam

Activities and deliverables

PORTAL	Study DMA and its API	Agree look with design team	Build bilingual RTL frontend	DELIVERABLE: Redesigned portal
ALERTS	Read monitoring results	Connect to SMS gateway	Add state and retries	Test with engineers	DELIVERABLE: SMS alert system

What I learned and what I take forward

TECHNICAL
Integrating with real external services, and working inside a legacy system I could not change or rebuild.

PRACTICAL
Seeing why a constraint like DMA's lockdown made the scope the right one, instead of accepting it at face value.

PROFESSIONAL
With real users, "working" means it helps them, not that it runs on my laptop.

BUILT ON MY DEGREE **Python and SQL** from my courses drove the alert script and database checks. **HTML and CSS** carried the portal. [confirm / add courses]

More detail and demo
Scan for the full write-up and screenshots

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